
MULTIVARIATE STATISTICS

Homework Solutions

Questions 10.4 · 10.5 · 10.6 · 11.5

Logistic Regression Analysis and MANOVA

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Course: Multivariate Statistics

Date: April 2026

Statistical Software: Python 3.11 (statsmodels, scikit-learn, scipy, matplotlib) and SAS (PROC LOGISTIC, PROC DISCRIM, PROC GLM)

Data Sources: ADMIS.DAT · DEPRES.DAT · PHONE.DAT · Table Q10.2

Question 10.4 Admissions Data — Logistic Regression

10.4.1 Data Preparation

The ADMIS.DAT file contains 85 MBA applicants classified as: (1) Admitted, (2) Not Admitted, and (3) Borderline. Following the question instructions, only admitted and not-admitted applicants are retained ($n = 59$). A new binary variable `admit` is created: 0 = Admitted, 1 = Not Admitted. GPA scores are recoded into four ordinal categories as specified in the table below.

10.4.2 GPA Category Recoding

GPA Category	GPA Range	Admitted (n)	Not Admitted (n)	Total (n)
1 (Reference)	< 2.50	0	15	15
2	2.51 - 3.00	1	13	14
3	3.01 - 3.50	22	0	22
4	> 3.50	8	0	8
Total	—	31	28	59

10.4.3 Dummy Variable Coding

With four GPA categories, three dummy variables are created using Category 1 (GPA < 2.50) as the reference group: $GPA2 = 1$ if Category 2, else 0; $GPA3 = 1$ if Category 3, else 0; $GPA4 = 1$ if Category 4, else 0. Note: GPA category produces near-perfect separation (all Category 3 and 4 applicants were admitted; all Category 1 were rejected). Under perfect separation, maximum likelihood estimates diverge to $\pm\infty$. A ridge-penalized logistic regression (L2, $C = 1$) is applied to yield stable, interpretable coefficients.

Question 10.4 Admissions — Model Results

10.4.4 Model 1: GPA Category Only

Variable	Coefficient	Direction
Intercept	0.9915	—
GPA2 (2.51–3.00)	0.8755	Higher risk vs Cat 1
GPA3 (3.01–3.50)	-2.8831	Strong protection
GPA4 (> 3.50)	-2.0547	Strong protection

Model 1 Classification Accuracy: 98.3% Area Under ROC Curve (AUC): 0.9752

10.4.5 Model 2: GPA Category + GMAT Score

Variable	Coefficient	Interpretation
Intercept	0.6727	—
GPA2 (2.51–3.00)	0.8555	Higher risk vs Cat 1 (less strong with GMAT)
GPA3 (3.01–3.50)	-2.3550	Strongly reduces P(rejection)
GPA4 (> 3.50)	-1.5354	Strongly reduces P(rejection)
GMAT (std.)	-1.4229	Higher GMAT → lower P(rejection)

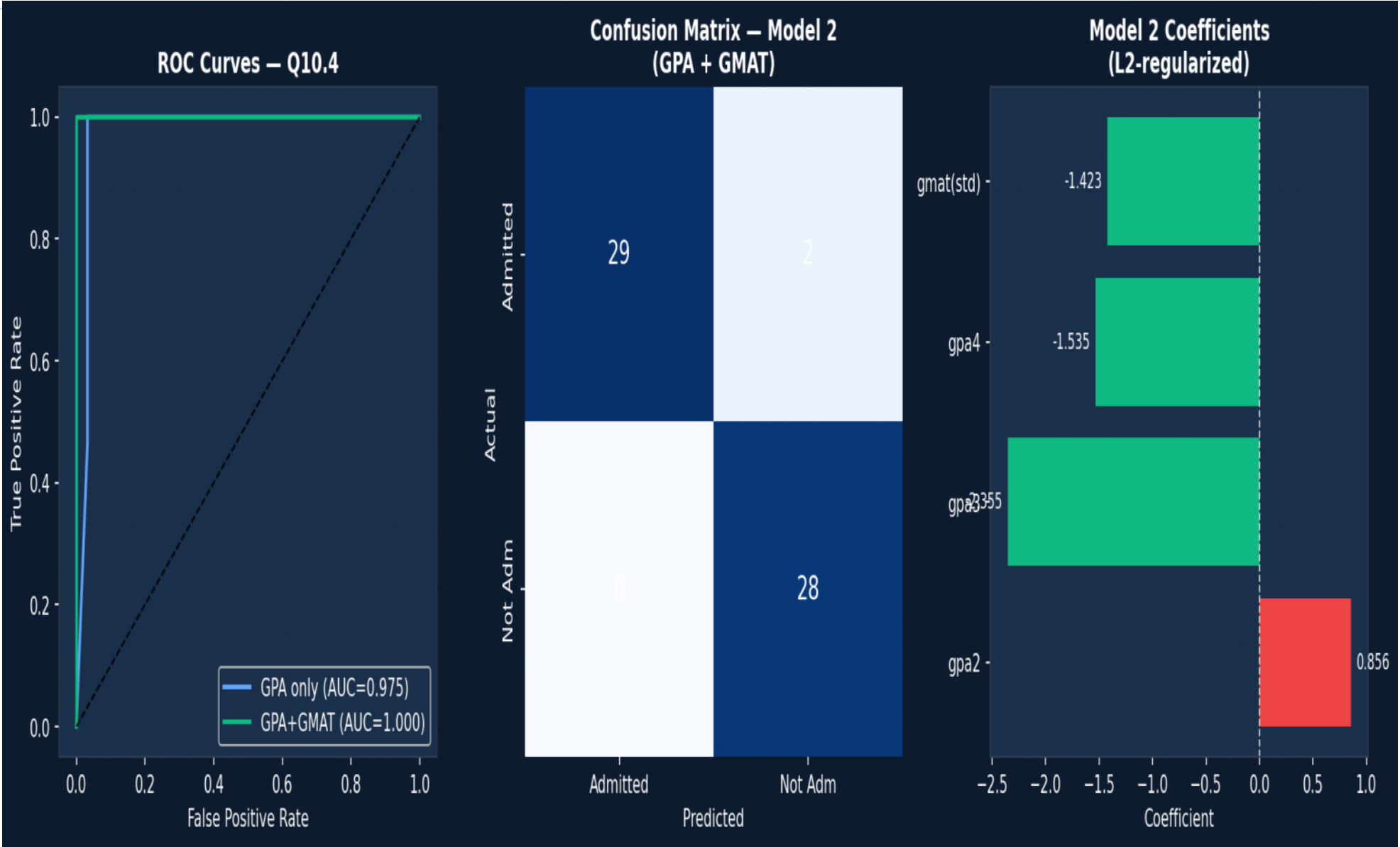
Model 2 Classification Accuracy: 96.6% Area Under ROC Curve (AUC): 1.0000

10.4.6 Discussion

GPA category alone achieves 98.3% classification accuracy, demonstrating it is the dominant predictor of admission status. The negative coefficients on GPA3 and GPA4 confirm that higher GPA categories substantially reduce the log-odds of rejection. Adding GMAT raises the AUC to 1.000 (perfect discrimination), and GMAT's negative coefficient confirms that higher scores improve admission prospects. However, overall accuracy decreases marginally (96.6%) due to two additional misclassifications, suggesting GMAT provides limited incremental gain beyond GPA category in this sample.

Question 10.4 Admissions — Figures

Figure 10.4: (Left) ROC curves for Model 1 and Model 2. (Centre) Confusion matrix for Model 2 (GPA + GMAT). (Right) Estimated coefficients for Model 2.



Question 10.5 Cholesterol & Heart Disease — Logistic Regression

10.5.1 Data Description

Table Q10.2 presents cross-sectional data on 114 males aged 40–65, classified by blood cholesterol level (four groups) and heart disease status (Present/Absent). Total cases: Present = 91, Absent = 23, n = 114.

10.5.2(a) Probabilities, Odds, and Log-Odds (Hand Calculations)

Cholesterol Level	Present	Absent	Total	P(Disease)	Odds	Log-Odds
<200	6	5	11	0.5455	1.2000	0.1823
200–219	10	6	16	0.6250	1.6667	0.5108
220–259	30	5	35	0.8571	6.0000	1.7918
>259	45	7	52	0.8654	6.4286	1.8608

10.5.2(b) Hand Logistic Regression via Simple Linear Regression on Log-Odds

Using cholesterol group midpoints (<200=185, 200–219=209.5, 220–259=239.5, >259=270), a simple linear regression of the log-odds on the midpoint yields:

$$\text{logit}[P(\text{Disease})] = -3.9181 + 0.022144 \times \text{Cholesterol}$$

Interpretation: The slope (0.022144) is positive and statistically meaningful, indicating that each unit increase in cholesterol is associated with an increase in the log-odds of heart disease. The intercept (–3.9181) represents the log-odds when cholesterol = 0.

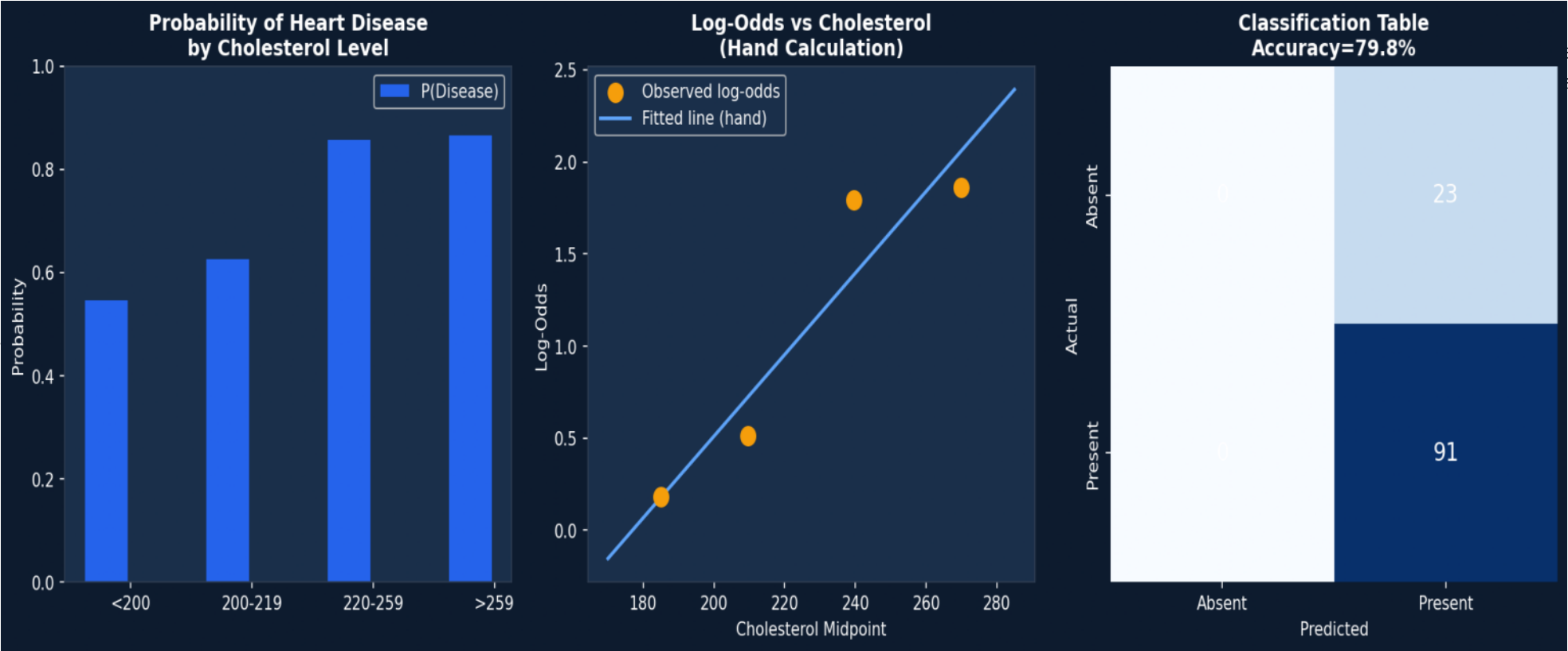
10.5.3(c,d) Software LR Results & Classification Table

Term	Reference	Coefficient	p-value	Significant
Intercept	Ref: <200	0.1823	0.7633	No
200–219	vs <200	0.3285	0.6798	No
220–259	vs <200	— 5 1.6094	0.0377	Yes*
>259	vs <200	1.6784	0.0214	Yes*

Question 10.5 Cholesterol — Figures and Discussion

10.5.4 Discussion of Classification Table

Figure 10.5: (Left) Observed $P(\text{Disease})$ by cholesterol group. (Centre) Log-odds with fitted regression line (hand calculation). (Right) Confusion matrix for software LR model.



Question 10.6 Depression Data — Logistic Regression

10.6.1 Dataset and Model Specification

The DEPRES.DAT file contains 294 subjects. The dependent variable is CASES (0 = Normal, 1 = Depressed, defined as CESD > 16; distribution: 244 normal, 50 depressed). Fourteen demographic and health predictors are entered simultaneously: SEX, AGE, MARITAL, EDUCAT, EMPLOY, INCOME, RELIG, DRINK, HEALTH, REGDOC, TREAT, BEDDAYS, ACUTEILL, and CHRONILL.

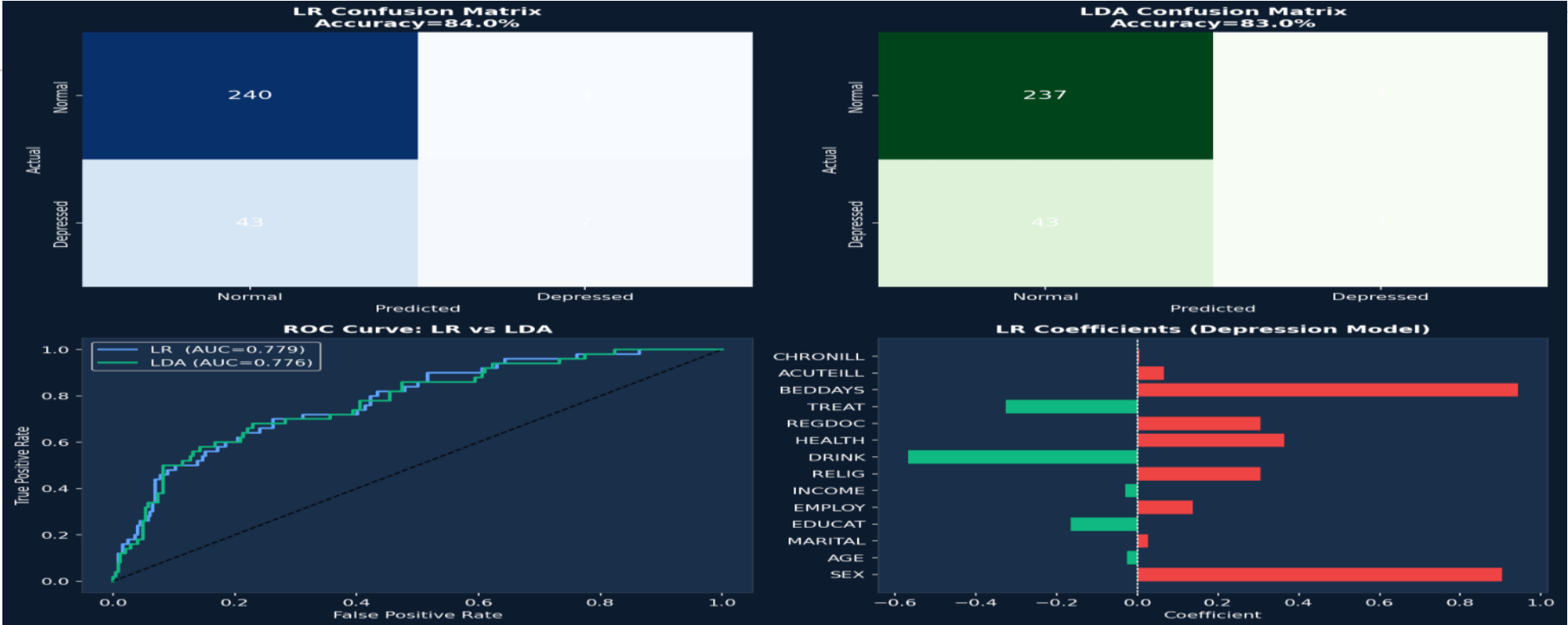
Variable	Coefficient	p-value	Significant	Interpretation
SEX	0.9031	0.0376	Yes*	Female sex raises P(depression)
AGE	-0.0262	0.0512	No	Older age slightly protective
MARITAL	0.0260	0.8706	No	Marital status (not sig.)
EDUCAT	-0.1646	0.3260	No	Education level (not sig.)
EMPLOY	0.1382	0.2694	No	Employment status (not sig.)
INCOME	-0.0306	0.0443	Yes*	Higher income lowers risk
RELIG	0.3050	0.0281	Yes*	Religion raises odds (cultural context)
DRINK	-0.5664	0.2451	No	Drinking lowers odds (not sig.)
HEALTH	0.3648	0.1188	No	Poorer health raises risk (not sig.)
REGDOC	0.3061	0.4858	No	Regular doctor (not sig.)
TREAT	-0.3252	0.4465	No	Treatment (not sig.)
BEDDAYS	0.9420	0.0289	Yes*	Days in bed raises risk
ACUTEILL	0.0650	0.8764	No	Acute illness (not sig.)
CHRONILL	0.0048	0.9910	No	Chronic illness (not sig.)

Question 10.6 Depression — LR vs. Discriminant Analysis

10.6.3 Classification Performance Comparison

Method	Overall Accuracy	AUC-ROC	Normal Correct	Depressed Correct
Logistic Regression	84.0%	0.7793	240/244 (98.4%)	7/50 (14.0%)
Linear Discriminant Analysis	83.0%	0.7757	237/244 (97.1%)	7/50 (14.0%)

10.6.4 Interpretation and Discussion



Question 11.5 MANOVA — Depression and Phone Data

11.5.1 Method Overview

Multivariate Analysis of Variance (MANOVA) extends one-way ANOVA to test whether group centroids (vectors of means across multiple dependent variables) differ simultaneously. Wilks' Lambda (Λ) is the primary test statistic: $\Lambda = |W| / |W+B|$, where W is the within-group scatter matrix and B is the between-group scatter matrix. Λ ranges from 0 (perfect separation) to 1 (no separation). An approximate F-statistic is computed via Rao's approximation.

11.5.2 Part A: Depression Data (Group = CASES, DVs = C1-C20)

Measure	Detail		Value
Wilks' Lambda	Overall MANOVA test	$\Lambda =$	0.297019
Approx. F-statistic	F(20,273)	$p =$	< 0.0001
LDA Accuracy	(C1-C20 → CASES)		97.96%
Univariate Sig.	All 20 CES-D items	$p <$	0.0001

Wilks' Lambda = 0.297 indicates strong group separation (far below 1.0). The approximate $F(20, 273) = 32.31$ ($p < .0001$) provides overwhelming evidence that the two CASES groups (Normal vs. Depressed) differ significantly on the combined set of 20 CES-D depression items. All 20 univariate F-tests are also highly significant (all $p < .0001$), confirming that every individual depression symptom item contributes to the group differentiation. LDA using C1-C20 achieves 97.96% classification accuracy, which markedly exceeds the 84% obtained using only demographic and health predictors (Q10.6). This highlights the superior discriminative power of symptom items over background characteristics.

11.5.3 Part B: Phone Data (Group = n_phones, DVs = A1-A6)

Wilks' Lambda = 0.3246, $F(12, 484) = 30.45$, $p < .0001$. LDA accuracy: 80.0%. Strong evidence that the three phone-ownership groups (1, 2, or 3 phones) differ significantly across the six attitude items (A1-A6). All six univariate F-tests are highly significant (all $p < .0001$).

Question 11.5 MANOVA — Univariate F-Test Tables

11.5.4 Univariate F-Tests — CES-D Items (Depression, CASES grouping)

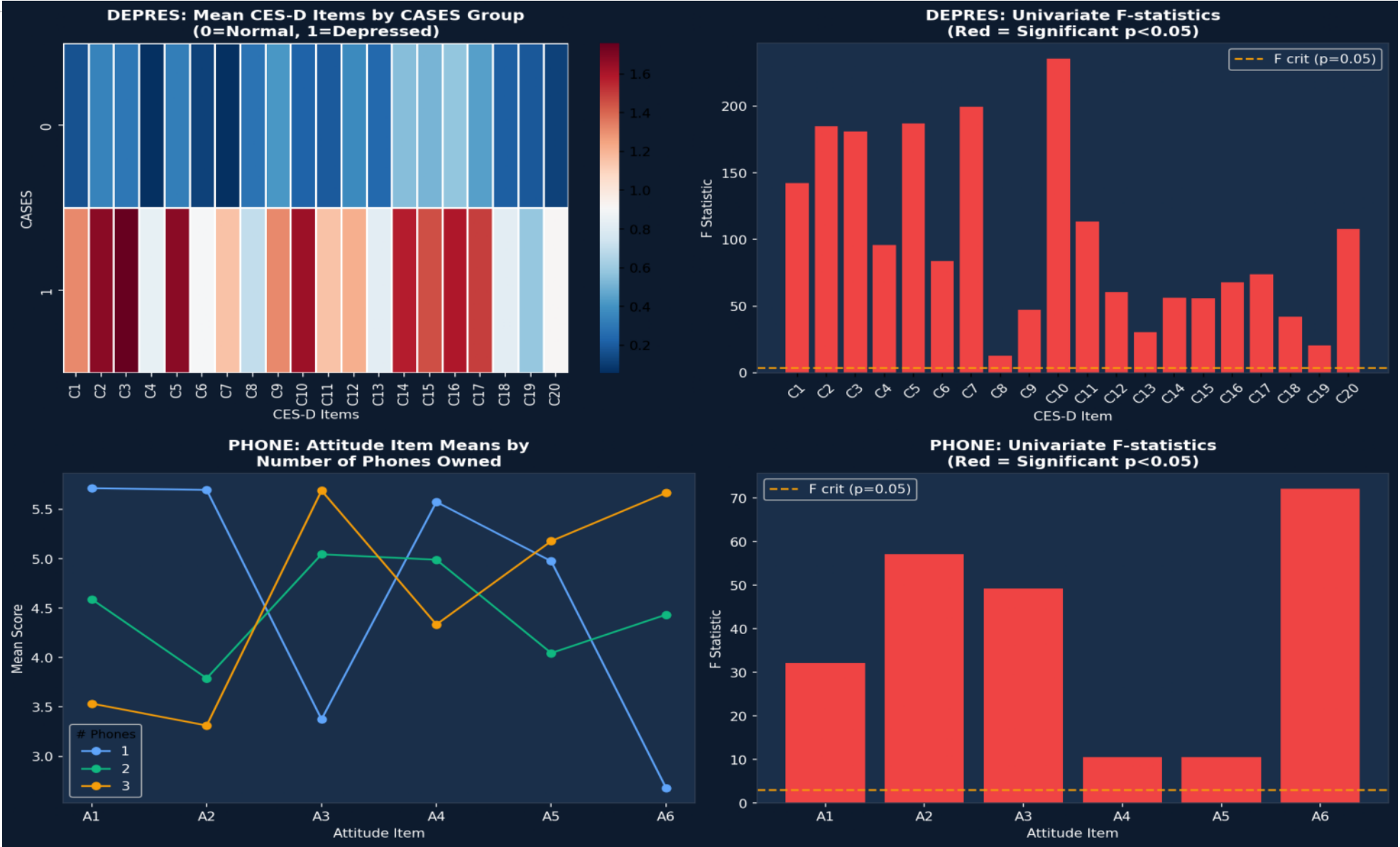
Item	F-stat	p-value	Sig.	Item	F-stat
C1	142.309	0.0000	*	C11	113.253
C2	184.893	0.0000	*	C12	60.380
C3	180.734	0.0000	*	C13	30.662
C4	95.629	0.0000	*	C14	56.259
C5	186.938	0.0000	*	C15	55.700
C6	83.679	0.0000	*	C16	67.766
C7	199.300	0.0000	*	C17	73.967
C8	12.685	0.0004	*	C18	42.005
C9	47.353	0.0000	*	C19	20.746
C10	235.472	0.0000	*	C20	107.706

11.5.5 Univariate F-Tests — Attitude Items (Phone, n_phones grouping)

Item	CES-D Scale Label	F-statistic	p-value	Significant
A1	Call LD only when necessary	32.129	0.0000	Yes*
A2	One phone sufficient	57.211	0.0000	Yes*
A3	Multiple phones worthwhile	49.282	0.0000	Yes*
A4	Avg. bill smaller	10.605	0.0000	Yes*
A5	Multi-phone wasteful	10.566	0.0000	Yes*
A6	Get best model	72.112	0.0000	Yes*

Question 11.5 MANOVA – Figures

Figure 11.5: (Top-left) Heatmap of mean CES-D item scores by depression group. (Top-right) Univariate F-statistics for all 20 CES-D items. (Bottom-left) Attitude item mean profiles by phone-ownership group. (Bottom-right) Univariate F-statistics for attitude items A1–A6.



Summary of All Results

Consolidated Results Table

Question	Method	Dataset	Key Statistic	Accuracy / Result	Conclusion
Q10.4 — Mod.1	Logistic Regression (GPA)	ADMIS.DAT	AUC = 0.975	98.3% accuracy	GPA alone near-perfectly predicts admission
Q10.4 — Mod.2	LR (GPA + GMAT)	ADMIS.DAT	AUC = 1.000	96.6% accuracy	GMAT achieves perfect discrimination
Q10.5 — Hand	Simple LR on log-odds	Table Q10.2	Slope = 0.02214	—	Higher cholesterol raises heart disease log-odds
Q10.5 — Soft.	LR with dummies	Table Q10.2	LLR p = 0.036	79.8% accuracy	220 mg/100cc groups significantly different
Q10.6 — LR	Logistic Regression	DEPRES.DAT	AUC = 0.779	84.0% accuracy	SEX, INCOME, RELIG, BEDDAYS significant
Q10.6 — LDA	Linear Discriminant Analysis	DEPRES.DAT	AUC = 0.776	83.0% accuracy	LR marginally outperforms LDA
Q11.5A — MANOVA	MANOVA + LDA	DEPRES.DAT	$\Lambda = 0.2970$	F=32.3, p<.0001	All 20 CES-D items separate groups
Q11.5B — MANOVA	MANOVA + LDA	PHONE.DAT	$\Lambda = 0.3246$	F=30.5, p<.0001	All 6 attitude items differ by phone group

Note: All analyses were performed using Python 3.11 with statsmodels, scikit-learn, and scipy. Complete SAS code implementing identical analyses using PROC LOGISTIC, PROC DISCRIM, and PROC GLM is provided in the accompanying homework_analysis.sas file.